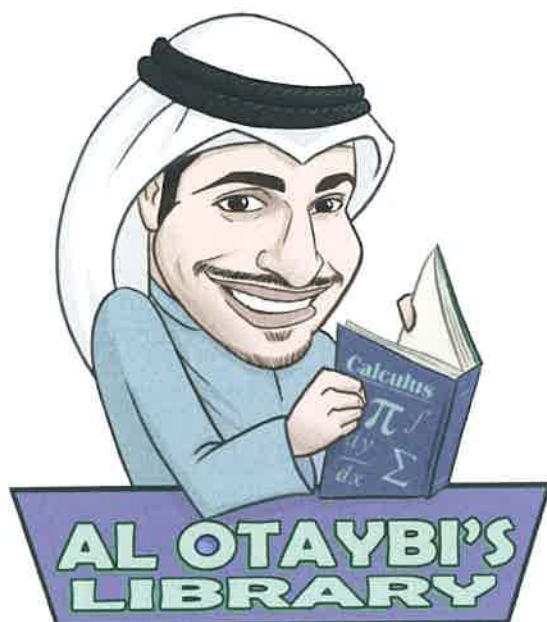


Kuwait University

2021/2022

Fall course



Calculus A

Second

Sections : 3.5 – 3.6

3.5 Implicit Differentiation.

3.6 Derivatives of Logarithmic Function.

(11)



3.5 The derivative of the inverse of trigonometric functions



Function (الدالة)	Derivative (المشتقة)
$\sin^{-1}x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1}x$	$\frac{-1}{\sqrt{1-x^2}}$
$\tan^{-1}x$	$\frac{1}{x^2+1}$
$\cot^{-1}x$	$\frac{-1}{x^2+1}$
$\sec^{-1}x$	$\frac{1}{x\sqrt{x^2-1}}$
$\csc^{-1}x$	$\frac{-1}{x\sqrt{x^2-1}}$



لازم تعرف اثباتاتهم

نهاية المذاكرة



للتسهيل



سالِب

يكون فيها



c

الدوال التي تبدأ بحرف



Big mistakes



~~$$f(x) = \sin^{-1}x \rightarrow f'(x) = \cos^{-1}x$$~~

~~$$\tan^{-1}x = \frac{\sin^{-1}x}{\cos^{-1}x}$$~~



Text book questions



Find the derivative of the function.

$$y = \tan^{-1}\sqrt{x}$$

$$y = \sin^{-1}(2x + 1)$$

$$y' = \frac{\frac{1}{2\sqrt{x}}}{(\sqrt{x})^2 + 1}$$

$$y' = \frac{2}{\sqrt{1 - (2x+1)^2}}$$

$$g(x) = \arccos\sqrt{x}$$

$$y = \tan^{-1}(x - \sqrt{1+x^2})$$

$$g(x) = \cos^{-1}\sqrt{x}$$

$$g'(x) = \frac{-\frac{1}{2\sqrt{x}}}{\sqrt{1 - (\sqrt{x})^2}}$$

$$y' = \frac{1 - \frac{2x}{2\sqrt{1+x^2}}}{(x - \sqrt{1+x^2})^2 + 1}$$

$$y = \sqrt{\tan^{-1}x}$$

$$y' = \frac{\frac{1}{x^2+1}}{2\sqrt{\tan^{-1}x}}$$

$$y = x\sin^{-1}x + \sqrt{1-x^2}$$

استق + استق

$$y' = (1)\sin^{-1}x + x \cdot \frac{1}{\sqrt{1-x^2}} + \frac{-2x}{2\sqrt{1-x^2}}$$

Find $f'(t)$ if

$$f(t) = \cot^{-1}(t) + \cot^{-1}\left(\frac{1}{t}\right)$$

$$f'(t) = \frac{-1}{t^2 + 1} + \frac{-\frac{1}{t^2}}{\left(\frac{1}{t}\right)^2 + 1}$$

مشتق $\cot^{-1}$ سالبة

مشتق $\frac{1}{t}$ سالبة

Find $f'(x)$ if

$$f(x) = (\arcsin 2x)^2$$

$$f(x) = (\sin^{-1} 2x)^2$$

$$f'(x) = 2 (\sin^{-1} 2x) \cdot \frac{2}{\sqrt{1 - (2x)^2}}$$

$$\arcsin x = \sin^{-1} x$$

Find y' if

$$y = \arctan(\arcsin \sqrt{x})$$

$$y = \tan^{-1}(\sin^{-1} \sqrt{x})$$

$$y' = \frac{1}{(\sin^{-1} \sqrt{x})^2 + 1} \cdot \frac{\frac{1}{2\sqrt{x}}}{\sqrt{1 - (\sqrt{x})^2}}$$

Find the derivative of the function

$$y = \arccos\left(\frac{b + a \cos x}{a + b \cos x}\right) \quad 0 \leq x \leq \pi, \quad a > b > 0$$

$$y = \cos^{-1}\left(\frac{b + a \cos x}{a + b \cos x}\right)$$

$$y' = \frac{-1}{\sqrt{1 - \left(\frac{b + a \cos x}{a + b \cos x}\right)^2}} \cdot \frac{(a + b \cos x)(-a \sin x) - (b + a \cos x)(-b \sin x)}{(a + b \cos x)^2}$$

3.5 Implicit Differentiation

Differentiate with respect to x

اشتقاق مباشر لكل x ثم يتم إضافة y' بعد كل اشتقاق
اشتقاق مباشر لكل y



ثم يتم جعل y' في طرف بروحها



Text book questions



Find dy/dx by implicit differentiation

$$x^2 - 4xy + y^2 = 4$$

Differentiate with respect to x

اشتقاق بشت + بشت اشتقا

$$2x - 4(1)y - 4x y' + 2y y' = 0$$

$$2y y' - 4x y' = 4y - 2x$$

$$y' [2y - 4x] = 4y - 2x$$

$$\div [2y - 4x]$$

$$y' = \frac{4y - 2x}{2y - 4x}$$



Text book questions



Find dy/dx by implicit differentiation

$$e^y \sin x = x + xy$$

Apr 2019

Diff. w.r. t. x

اشتق بـ y + مشتق بـ x اشتق بـ y + مشتق بـ x

$$e^y y' \sin x + e^y \cos x = 1 + (1)y + x y'$$

$$e^y y' \sin x - x y' = 1 + y - e^y \cos x$$

$$y' [e^y \sin x - x] = 1 + y - e^y \cos x$$

$$\div [e^y \sin x - x]$$

$$y' = \frac{1 + y - e^y \cos x}{e^y \sin x - x}$$



Text book questions



Find dy/dx by implicit differentiation

$$e^{\frac{x}{y}} = x - y$$

Diff. w.r.t. x

$$e^{\frac{x}{y}} \cdot \frac{y(1) - xy'}{y^2} = 1 - y'$$

* y^2

$$e^{\frac{x}{y}} [y - xy'] = y^2 - y^2 y'$$

$$ye^{\frac{x}{y}} - xe^{\frac{x}{y}} y' = y^2 - y^2 y'$$

$$y^2 y' - xe^{\frac{x}{y}} y' = y^2 - ye^{\frac{x}{y}}$$

$$y' [y^2 - xe^{\frac{x}{y}}] = y^2 - ye^{\frac{x}{y}}$$

$$\div [y^2 - xe^{\frac{x}{y}}]$$

$$y' = \frac{y^2 - ye^{\frac{x}{y}}}{y^2 - xe^{\frac{x}{y}}}$$



Text book questions



Find dy/dx by implicit differentiation

$$\frac{x^2}{x+y} = y^2 + 1$$

Diff. w.r.t. x

$$\frac{(x+y) 2x - x^2(1+y')}{(x+y)^2} = 2y y'$$

* $(x+y)^2$

$$(x+y) 2x - x^2(1+y') = 2y y' (x+y)^2$$

$$\underline{2x^2} + 2xy - \underline{x^2} - x^2 y' = 2y y' (x+y)^2$$

$$2y y' (x+y)^2 + x^2 y' = x^2 + 2xy$$

$$y' [2y(x+y)^2 + x^2] = x^2 + 2xy$$

$$\div [2y(x+y)^2 + x^2]$$

$$y' = \frac{x^2 + 2xy}{2y(x+y)^2 + x^2}$$



Text book questions



Find dy/dx by implicit differentiation

$$\sin(xy) = \cos(x + y)$$

Diff. w.r.t. x

اشتق بشأ + مشتق الاشتق

$$\cos(xy) [(1) y + x y'] = -\sin(x+y) [1 + y']$$

$$\cos(xy) \cdot y + x \cos(xy) y' = -\sin(x+y) - \sin(x+y) y'$$

$$x \cos(xy) y' + \sin(x+y) y' = -\sin(x+y) - \cos(xy) y$$

$$y' [x \cos(xy) + \sin(x+y)] = -\sin(x+y) - \cos(xy) y$$

$$\div [x \cos(xy) + \sin(x+y)]$$

$$y' = \frac{-\sin(x+y) - \cos(xy) y}{x \cos(xy) + \sin(x+y)}$$



Text book question



Find dy/dx by implicit differentiation

$$\tan^{-1}(x^2y) = x + xy^2$$

Diff. w.r.t. x

اشتق بشتا + بشتا اشتق

اشتق بشتا + بشتا اشتق

$$\frac{2xy + x^2 y'}{(x^2y)^2 + 1} = 1 + (1)y^2 + x 2y y'$$

$$* [(x^2y)^2 + 1]$$

$$2xy + x^2 y' = (x^2y)^2 + 1 + y^2 [(x^2y)^2 + 1] + 2xy y' [(x^2y)^2 + 1]$$

$$x^2 y' - 2xy y' [(x^2y)^2 + 1] = (x^2y)^2 + 1 + y^2 [(x^2y)^2 + 1] - 2xy$$

$$y' [x^2 - 2xy [(x^2y)^2 + 1]] = (x^2y)^2 + 1 + y^2 [(x^2y)^2 + 1] - 2xy$$

$$y' = \frac{(x^2y)^2 + 1 + y^2 [(x^2y)^2 + 1] - 2xy}{x^2 - 2xy [(x^2y)^2 + 1]}$$

Let $x = \frac{x + xy + 2}{y^2}$. Find $\frac{dy}{dx}$ at $y = 1$

Past exam

$$\text{at } y=1 \longrightarrow x = \frac{x + x + 2}{1}$$

$$x = 2x + 2$$

$$2x - x = -2 \longrightarrow x = -2$$

$$\Rightarrow P(-2, 1)$$

Differentiate w.r.t. x

$$1 = \frac{y^2 [1 + (1)y + x y'] - (x + xy + 2) 2y y'}{(y^2)^2}$$

$$\text{at } P(-2, 1)$$

$$1 = \frac{(1) [1 + 1 - 2 y'] - (-2 - 2 + 2) 2(1) y'}{1}$$

$$1 = \frac{2 - 2 y' + 2 (2 y')}{1}$$

$$1 = 2 - 2 y' + 4 y'$$

$$-1 = 2 y'$$

$$y' = -\frac{1}{2}$$



Tricky question !!



Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 0$, where

$$2y + \sin(xy) = 1$$

at $x=0 \rightarrow 2y + \sin 0 = 1 \xrightarrow{\div 2} y = \frac{1}{2} \rightarrow P(0, \frac{1}{2})$

Diff. w.r.t. x

اشتق بشت + بشت اشتقا
 $2y' + \cos(xy) [(1)y + x y'] = 0 \implies (1)$

at $P(0, \frac{1}{2})$

$$2y' + \cos(0) [\frac{1}{2} + 0] = 0$$

$$2y' + (1) \frac{1}{2} = 0$$

$$2y' = -\frac{1}{2} \xrightarrow{\times \frac{1}{2}} \frac{1}{2} 2y' = -\frac{1}{2} \frac{1}{2} \rightarrow y' = -\frac{1}{4}$$

~~$y'' = 0$~~

الاشتقاق
قبل التعويض



Diff. (1) w.r.t. x

اشتق بشت + بشت اشتق

$$2y'' - \sin(xy) [(1)y + x y'] [y + x y'] + \cos(xy) [y' + (1)y + x y''] = 0$$

at $P(0, \frac{1}{2}), y' = -\frac{1}{4}$

$$2y'' - 0 + \cos(0) [\frac{-1}{4} + \frac{-1}{4} + 0] = 0$$

$$2y'' + (1) [\frac{-2}{4}] = 0$$

$$2y'' = \frac{1}{2} \xrightarrow{\times \frac{1}{2}} \frac{1}{2} 2y'' = \frac{1}{2} \frac{1}{2} \rightarrow y'' = \frac{1}{4}$$



Text book question



If $f(x) + x^2[f(x)]^3 = 10$ and $f(1) = 2$, find $f'(1)$.

Diff. w.r.t. x

الشتق بشتا + بشتا اشتقاق

$$f'(x) + 2x [f(x)]^3 + x^2 \cdot 3 [f(x)]^2 f'(x) = 0$$

at $x=1$

$$f'(1) + 2(1) [f(1)]^3 + (1) \cdot 3 [f(1)]^2 f'(1) = 0$$

$$f'(1) + 2 [2]^3 + 3 [2]^2 f'(1) = 0$$

$$f'(1) + 2(8) + 12 f'(1) = 0$$

$$13 f'(1) = -16$$

$$f'(1) = \frac{-16}{13}$$



Text book question



If $g(x) + x \sin g(x) = x^2$, find $g'(0)$.

at $x=0$

$$g(0) + 0 = 0 \longrightarrow \boxed{g(0) = 0}$$

Differentiate w.r. to x

اشتقاق بشت + مشتق

$$g'(x) + (1) \sin g(x) + x \cos g(x) \cdot g'(x) = 2x$$

at $x=0$

$$g'(0) + \sin g(0) + 0 = 0$$

$$g'(0) + \sin 0 = 0$$

$$g'(0) = 0$$

$$\text{If } f(x) + x \sin(f(x)) = x^3 + \frac{\pi}{2}$$

Nov 2018

then $f'(0)$ is equal to :

a) 2 b) -1 c) 0 d) 1 e) None of the above

at $x=0$

$$f(0) + 0 = 0 + \frac{\pi}{2} \longrightarrow$$

$$f(0) = \frac{\pi}{2}$$

$$f(x) + x \sin(f(x)) = x^3 + \frac{\pi}{2}$$

Diff. w.r.t. x

اشتقاق بشي + مشتق اشتقا

$$f'(x) + (1) \sin(f(x)) + x \cos(f(x)) f'(x) = 3x^2$$

at $x=0$

$$f'(0) + \sin(f(0)) + 0 = 0$$

$$f'(0) + \sin \frac{\pi}{2} = 0$$

$$f'(0) + 1 = 0$$

$$f'(0) = -1$$

(b) ✓

If $y = x^2 + x \sin y + \frac{\pi}{2}$, at $x = 0$ Find

a) Equation of the tangent line.

b) Equation of the normal line.

Past exam

$$\text{at } x=0 \longrightarrow y = 0 + 0 + \frac{\pi}{2} \longrightarrow P(0, \frac{\pi}{2})$$

Diff. w.r.t. x

$$y' = 2x + (1) \sin y + x \cos y \cdot y'$$

at $P(0, \frac{\pi}{2})$

$$y' = 0 + \sin \frac{\pi}{2} + 0 \longrightarrow y' = 1$$

a) Equation of the Tangent Line :

$$y - y_0 = m_t (x - x_0)$$

$$y - \frac{\pi}{2} = 1 \cdot (x - 0)$$

b) Equation of the Normal Line :

$$y - y_0 = m_n (x - x_0)$$

$$y - \frac{\pi}{2} = -\frac{1}{1} (x - 0)$$

Find an equation of the tangent line of

$$(x + y)^3 = 1 + x^2 e^y$$

at the point whose x-coordinate is 0

Past exam

at $x=0 \longrightarrow (0+y)^3 = 1+0 \longrightarrow y^3=1 \longrightarrow y=1$
 $\Rightarrow P(0,1)$

Diff. w.r.t. x

الشتق بـ y + شتق استفا
 $3(x+y)^2(1+y') = 0 + 2xe^y + x^2 e^y y'$

at $P(0,1)$

$$3(0+1)^2(1+y') = 0 + 0$$

$$3(1)(1+y') = 0$$

$$3 + 3y' = 0$$

$$3y' = -3 \xrightarrow{\div 3} y' = -1$$

∴ The equation of the tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - 1 = -1(x - 0)$$

Find an equation of the normal line to

$$\sec(x^2y) + \sqrt{x^2 + y} - x = 3$$

at $x = 0$.

Past exam

$$\text{at } x=0 \longrightarrow \sec 0 + \sqrt{0+y} - 0 = 3$$

$$\frac{1}{\cos 0} + \sqrt{y} = 3$$

$$\frac{1}{1} + \sqrt{y} = 3$$

$$\sqrt{y} = 2 \longrightarrow y = 4 \longrightarrow P(0,4)$$

Diff. w.r.t. x

$$\sec(x^2y) \tan(x^2y) [2xy + x^2 y'] + \frac{2x + y'}{2\sqrt{x^2+y}} - 1 = 0$$

at $P(0,4)$

$$0 + \frac{0 + y'}{2\sqrt{0+4}} - 1 = 0$$

$$\frac{y'}{2(2)} = 1 \longrightarrow y' = 4$$

∴ The equation of the normal Line :

$$y - y_0 = m(x - x_0)$$

$$y - 4 = -\frac{1}{4}(x - 0)$$

Find the equation of the tangent lines to the curve

$$5y^2 - 4xy^3 = x^2 \quad \text{at } y = 1$$

Nov 2018

$$\text{at } y=1 \longrightarrow 5(1)^2 - 4x(1)^3 = x^2$$

$$5 - 4x = x^2$$

$$x^2 + 4x - 5 = 0$$

$$(x - 1)(x + 5) = 0$$

$$x = 1 \quad x = -5$$

$$\implies P(1, 1), P(-5, 1)$$

Diff. w.r. to x

$$10y y' - 4(1) y^3 - 4x 3y^2 y' = 2x$$

at $P(1, 1)$

$$10(1)y' - 4(1) - 4(3)y' = 2$$

$$10y' - 12y' = 6$$

$$-2y' = 6$$

$\div (-2)$

$$y' = -3$$

at $P(-5, 1)$

$$10(1)y' - 4 - 4(5)(3)(1)y' = -10$$

$$10y' - 4 + 60y' = -10$$

$$70y' = -6$$

$\div (70)$

$$y' = \frac{-6}{70}$$

$$y' = \frac{-3}{35}$$

$\therefore$ The equation of tangent line(s) :

$$y - y_0 = m(x - x_0)$$

$$y - 1 = -3(x - 1)$$

$$y - y_0 = m(x - x_0)$$

$$y - 1 = \frac{-3}{35}(x - (-5))$$

Show that the curves $x^2 + y^2 = r^2$ and $y = mx$ are orthogonal (meet at right angles) for all constants $m \neq 0, r \neq 0$

Past exam

$$x^2 + y^2 = r^2$$

Diff. w.r.t. x

$$2x + 2yy' = 0$$

$$2yy' = -2x$$

$\div 2y$

$$y' = \frac{-2x}{2y}$$

$$y'_1 = \frac{-x}{y}$$

$$y'_1 = \frac{-x}{mx}$$

$$y'_1 = \frac{-1}{m}$$

$$y = mx$$

$$y'_2 = m$$

$$\therefore y'_1 = \frac{-1}{y'_2}$$

$\therefore$ The curves are orthogonal.

If $f(x) = \sin^{-1}(2x)$, then $f'(x) =$

July 2017

- a) $\frac{1}{\sqrt{1-4x^2}}$ b) $\frac{-1}{\sqrt{1-4x^2}}$ c) $\frac{2}{\sqrt{1-4x^2}}$
d) $\frac{-2}{\sqrt{1-4x^2}}$ e) None of the above

$$f'(x) = \frac{2}{\sqrt{1-(2x)^2}}$$
$$= \frac{2}{\sqrt{1-4x^2}} \quad (c)$$

The slope of the tangent line to the graph of $x^2 - 3y^3 = -2$ at $(1, 1)$ is :

- a) $1/3$ b) $-9/2$ c) $2/9$
d) $3/2$ e) None of the above

July 2017

Diff. w.r.t. x

$$2x - 9y^2 y' = 0$$

at $(1, 1)$

$$2(1) - 9(1) y' = 0$$

$$9y' = 2 \xrightarrow{\div 9} y' = \frac{2}{9} \quad (c)$$

If $\sin x + y^3 = x^2 + 1$, then $\frac{dy}{dx} \Big|_{(0,1)} =$

May 2017

a) $-1/3$ b) $1/3$ c) 0 d) 1 e) None of the above

Diff. w.r.t. x

$$\cos x + 3y^2 y' = 2x$$

at (0,1)

$$\cos(0) + 3(1)y' = 0$$

$$1 = -3y'$$

$$y' = -\frac{1}{3} \quad (a)$$

If $f(x) = 5x$ and $g(x) = 2\cos x$, then $(f \circ g)'(\pi) =$

a) -5 b) 0 c) 10 d) -10 e) None of the above

May 2017

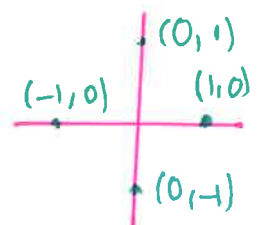
$$f \circ g = 5(2\cos x) = 10\cos x$$

$$(f \circ g)' = -10\sin x$$

at $x = \pi$

$$(f \circ g)'(\pi) = -10\sin \pi = 0 \quad (b)$$

$(\cos, \sin)$





If $y = \sin^{-1}(x)$ Show that

$$y' = \frac{1}{\sqrt{1-x^2}}$$

Let $y = \sin^{-1} x$

$$\sin y = \sin \sin^{-1} x$$

$$\sin y = x$$

Differentiate both sides w.r.t. x

$$\cos y \cdot y' = 1$$

$$y' = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

$$\begin{aligned}\cos^2 y + \sin^2 y &= 1 \\ \cos^2 y &= 1 - \sin^2 y \\ \cos y &= \sqrt{1 - \sin^2 y}\end{aligned}$$

If $y = \cos^{-1}(2x)$ Show that

$$y' = \frac{-2}{\sqrt{1-4x^2}}$$

Let $y = \cos^{-1}(2x)$

$$\cos y = \cos \cos^{-1}(2x)$$

$$\cos y = 2x$$

Differentiate both sides w.r.t. x

$$-\sin y \cdot y' = 2$$

$$\begin{aligned}y' &= \frac{2}{-\sin y} \\ &= \frac{2}{-\sqrt{1-\cos^2 y}} = \frac{-2}{\sqrt{1-(2x)^2}} \\ &= \frac{-2}{\sqrt{1-4x^2}}\end{aligned}$$

$$\begin{aligned}\cos^2 y + \sin^2 y &= 1 \\ \sin^2 y &= 1 - \cos^2 y \\ \sin y &= \sqrt{1 - \cos^2 y}\end{aligned}$$

If $y = \sec^{-1}(x)$ Show that

$$y' = \frac{1}{x\sqrt{x^2 - 1}}$$

Let $y = \sec^{-1} x$

$$\sec y = \sec \sec^{-1} x$$

$$\sec y = x$$

Differentiate both sides w.r.t. x

$$\sec y \tan y \cdot y' = 1$$

$$y' = \frac{1}{\sec y \tan y} \longrightarrow$$
$$= \frac{1}{\sec y \sqrt{\sec^2 y - 1}} = \frac{1}{x\sqrt{x^2 - 1}}$$

$$1 + \tan^2 y = \sec^2 y$$
$$\tan^2 y = \sec^2 y - 1$$
$$\tan y = \sqrt{\sec^2 y - 1}$$

If $y = \csc^{-1}(x)$ Show that

$$y' = \frac{-1}{x\sqrt{x^2 - 1}}$$

Let $y = \csc^{-1} x$

$$\csc y = \csc \csc^{-1} x$$

$$\csc y = x$$

Differentiate both sides w.r.t. x

$$-\csc y \cot y \cdot y' = 1$$

$$y' = \frac{1}{-\csc y \cot y} \longrightarrow$$
$$= \frac{1}{-\csc y \sqrt{\csc^2 y - 1}}$$
$$= \frac{-1}{x\sqrt{x^2 - 1}}$$

$$\cot^2 y + 1 = \csc^2 y$$
$$\cot^2 y = \csc^2 y - 1$$
$$\cot y = \sqrt{\csc^2 y - 1}$$

If $y = \tan^{-1}(x)$ Show that

$$y' = \frac{1}{x^2 + 1}$$

Let $y = \tan^{-1} x$

$$\tan y = \tan \tan^{-1} x$$

$$\tan y = x$$

Differentiate both sides w.r.t. x

$$\sec^2 y \cdot y' = 1$$

$$y' = \frac{1}{\sec^2 y}$$

$$= \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$$

$$1 + \tan^2 y = \sec^2 y$$

If $y = \cot^{-1}(x)$ Show that

$$y' = \frac{-1}{x^2 + 1}$$

Let $y = \cot^{-1} x$

$$\cot y = \cot \cot^{-1} x$$

$$\cot y = x$$

Differentiate both sides w.r.t. x

$$-\csc^2 y \cdot y' = 1$$

$$y' = \frac{1}{-\csc^2 y}$$

$$= \frac{-1}{\cot^2 y + 1}$$

$$= \frac{-1}{x^2 + 1}$$

$$\cot^2 y + 1 = \csc^2 y$$



3.6 Derivative of logarithmic function



Function (الدالة)	Derivative (المشتقة)
$\ln x$	$\frac{1}{x}$
$\ln f(x)$	$\frac{f'(x)}{f(x)}$



نصيحة



عند امكانية تبسيط $\ln$ بتطبيق خواصها فانه يفضل ذلك قبل الاشتقاق

$$\ln(xy) = \ln x + \ln y$$

$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$$

$$\ln(x)^y = y \ln x$$

$$\ln e^x = x$$

Find y'

$$y = \ln x + \ln(x^2 + 1) - \ln(\sin x + 3)$$

$$y' = \frac{1}{x} + \frac{2x}{x^2 + 1} - \frac{\cos x}{\sin x + 3}$$

Find y'

$$y = \ln x^7 + \ln\left(\frac{\sin x}{e^x}\right) + \ln(x \cos x)$$

$$y = 7 \ln x + \ln \sin x - \ln e^x + \ln x + \ln \cos x$$

$$y = 7 \ln x + \ln \sin x - x + \ln x + \ln \cos x$$

$$y' = 7 \frac{1}{x} + \frac{\cos x}{\sin x} - 1 + \frac{1}{x} + \frac{-\sin x}{\cos x}$$

Find $g'(x)$

$$g(x) = \ln(x\sqrt{x^2 - 1})$$

$$g(x) = \ln x + \ln(x^2 - 1)^{\frac{1}{2}}$$

$$g(x) = \ln x + \frac{1}{2} \ln(x^2 - 1)$$

$$g'(x) = \frac{1}{x} + \frac{1}{2} \cdot \frac{2x}{x^2 - 1}$$

Find $g'(y)$

$$g(y) = \ln \frac{(2y + 1)^5}{\sqrt{y^2 + 1}}$$

$$g(y) = \ln(2y + 1)^5 - \ln(y^2 + 1)^{\frac{1}{2}}$$

$$g(y) = 5 \ln(2y + 1) - \frac{1}{2} \ln(y^2 + 1)$$

$$g'(y) = 5 \frac{2}{2y + 1} - \frac{1}{2} \frac{2y}{y^2 + 1}$$

Find y'

$$y = \ln \frac{(x + 1)^5 \sin^2 x}{\sqrt{x + 1} e^x}$$

$$y = \ln(x + 1)^5 + \ln \sin^2 x - \ln(x + 1)^{\frac{1}{2}} - \ln e^x$$

$$y = 5 \ln(x + 1) + 2 \ln \sin x - \frac{1}{2} \ln(x + 1) - x$$

$$y' = 5 \frac{1}{x + 1} + 2 \frac{\cos x}{\sin x} - \frac{1}{2} \frac{1}{x + 1} - 1$$



Text book questions



Differentiate the function

$$y = \ln|1 + t - t^2|$$

$$y' = \frac{1-2t}{1+t-t^2}$$

$$y = \sin(\ln x)$$

$$y' = \cos(\ln x) \cdot \frac{1}{x}$$

$y = \tan[\ln(ax + b)]$, where a and b are constants

$$y' = \sec^2[\ln(ax+b)] \cdot \frac{a}{ax+b}$$

$$f(x) = x \ln x - x$$

اشتقاق x + مشتق $\ln x$

$$f'(x) = (1) \ln x + x \left(\frac{1}{x} \right) - 1$$

$$= \ln x$$

$$F(t) = (\ln t)^2 \sin t$$

اشتقاق $\sin t$ + مشتق $(\ln t)^2$

$$F'(t) = 2(\ln t) \cdot \frac{1}{t} \sin t + (\ln t)^2 \cos t$$

$$y = \ln(e^{-x} + xe^{-x})$$

اشتقاق e^{-x} + مشتق xe^{-x}

$$y' = \frac{e^{-x}(-1) + (1)e^{-x} + xe^{-x}(-1)}{e^{-x} + xe^{-x}}$$

Find y'' if $y = \ln(1 + \ln x)$

Apr 2019

$$y' = \frac{\frac{1}{x}}{1 + \ln x}$$

$$y' = \frac{1}{x(1 + \ln x)}$$

$$\begin{aligned} y'' &= \frac{[x(1 + \ln x)](1)' - 1 \cdot [x(1 + \ln x)]'}{[x(1 + \ln x)]^2} \\ &= \frac{[x(1 + \ln x)] \cdot 0 - [(1)(1 + \ln x) + x \cdot \frac{1}{x}]}{[x(1 + \ln x)]^2} \end{aligned}$$

الشفق بشفق + بشفق الشفق

$$= \frac{-[1 + \ln x + 1]}{[x(1 + \ln x)]^2}$$



Text book question



Differentiate f and find the domain of f

$$f(x) = \frac{x}{1 - \ln(x - 1)}$$

$$f'(x) = \frac{[1 - \ln(x - 1)](1) - x \cdot \frac{-1}{x - 1}}{[1 - \ln(x - 1)]^2}$$

Domain of $f(x)$

$$x - 1 > 0$$

$$x > 1$$

$$(1, \infty)$$

$$1 - \ln(x - 1) = 0$$

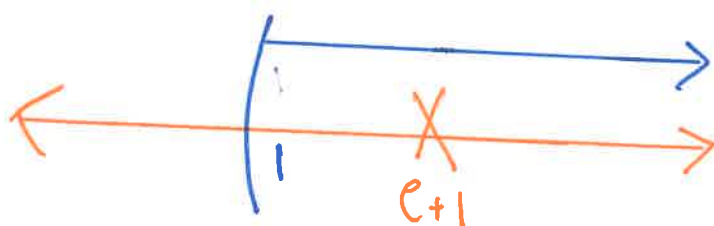
$$1 = \ln(x - 1)$$

$$e^1 = e^{\ln(x - 1)}$$

$$e^1 = x - 1$$

$$x = e + 1$$

$$\mathbb{R} \setminus \{e + 1\}$$



$$\text{Domain of } f(x) = (1, \infty) \setminus \{e + 1\}$$



Text book question



Find the equation of the tangent line of

$$y = x^2 \ln x, \text{ at } x = 1$$

at $x = 1 \longrightarrow y = (1) \ln 1 = 0 \longrightarrow P(1, 0)$

استقفا بـ x + بـ y استقفا

$$y' = 2x \ln x + x^2 \cdot \frac{1}{x}$$

at $(1, 0)$

$$y' = 2(1) \ln 1 + 1 \cdot \frac{1}{1}$$

$$y' = 0 + 1 \longrightarrow y' = 1$$

∴ The equation of the tangent line:

$$y - y_0 = m(x - x_0)$$

$$y - 0 = 1 \cdot (x - 1)$$



Remark



$$\log_a x = \frac{1}{\ln a} \ln x$$

$$\log_5 x \Rightarrow \frac{1}{\ln 5} \ln x$$

$$\log_x 3 \Rightarrow \frac{1}{\ln x} \ln 3$$

$$\log x \Rightarrow \frac{1}{\ln 10} \ln x$$

Find y'

$$y = \log_7(x^3 + 1)$$

$$y = \frac{1}{\ln 7} \ln(x^3 + 1)$$

$$y' = \frac{1}{\ln 7} \frac{3x^2}{x^3 + 1}$$

Find y'

$$y = \log_x(\sin x - 1)$$

$$y = \frac{1}{\ln x} \ln(\sin x - 1)$$

$$y = \frac{\ln(\sin x - 1)}{\ln x}$$

$$y' = \frac{(\ln x) \frac{\cos x}{\sin x - 1} - \ln(\sin x - 1) \frac{1}{x}}{(\ln x)^2}$$

Find y'

$$y = \log_4 3^x$$

$$y = \frac{1}{\ln 4} \cdot \ln 3^x = \frac{1}{\ln 4} x \cdot \ln 3$$

$$y = \frac{\ln 3}{\ln 4} x$$

$$\Rightarrow y' = \frac{\ln 3}{\ln 4} (1)$$

Find $f'(x)$

$$f(x) = \log_3 (3x^2 + 2)^7$$

$$f(x) = \frac{1}{\ln 3} \ln (3x^2 + 2)^7$$
$$= \frac{7}{\ln 3} \ln (3x^2 + 2)$$

$$\Rightarrow f'(x) = \frac{7}{\ln 3} \cdot \frac{6x}{3x^2 + 2}$$

Find y'

$$y = \log_5 (xe^x)$$

$$y = \frac{1}{\ln 5} \ln (xe^x)$$

$$= \frac{1}{\ln 5} [\ln x + \ln e^x]$$

$$= \frac{1}{\ln 5} [\ln x + x]$$

$$\Rightarrow y' = \frac{1}{\ln 5} \left[\frac{1}{x} + 1 \right]$$

Find y'

$$y = \log_{\sin x}(x)$$

$$y = \frac{1}{\ln \sin x} \ln x$$

$$y = \frac{\ln x}{\ln \sin x}$$

$$y' = \frac{\ln \sin x \cdot \frac{1}{x} - \ln x \cdot \frac{\cos x}{\sin x}}{(\ln \sin x)^2}$$



Tricky question !!



Find the constant a if the derivative of

$$\log_a x \text{ is } \frac{3}{x}$$

Let $y = \log_a x$

$$y = \frac{1}{\ln a} \ln x$$

$$y' = \frac{1}{\ln a} \cdot \frac{1}{x}$$

$$\Rightarrow \because y' = \frac{3}{x}$$

$$\therefore \frac{1}{\ln a} \cdot \frac{1}{x} = \frac{3}{x}$$

$$x = 3 \ln a \cdot x$$

$$\frac{x}{3x} = \ln a$$

$$\div 3x$$

$$\ln a = \frac{1}{3}$$

$$e^{\ln a} = e^{\frac{1}{3}}$$

$$a = e^{\frac{1}{3}}$$

Let $f(x) = \log_a(2x^3 - 1)$

Nov 2018

For what value of a , if any, is $f'(1) = 3$

$$f(x) = \frac{1}{\ln a} \cdot \ln(2x^3 - 1)$$

$$f'(x) = \frac{1}{\ln a} \cdot \frac{6x^2}{2x^3 - 1}$$

$$\begin{aligned} f'(1) &= \frac{1}{\ln a} \cdot \frac{6}{2-1} \\ &= \frac{1}{\ln a} \cdot \frac{6}{1} \\ &= \frac{1}{\ln a} \cdot 6 \end{aligned}$$

$$\therefore f'(1) = 3$$

$$\frac{1}{\ln a} \cdot 6 = 3$$

$\times \ln a$

$$6 = 3 \ln a$$

$\div 3$

$$\ln a = \frac{6}{3}$$

$$\ln a = 2$$

$$\cancel{e^{\ln a}} = e^2$$

$$a = e^2$$

Find y'

$$y = 2x \log_{10} \sqrt{x}$$

$$y = 2x \frac{1}{\ln 10} \ln \sqrt{x}$$

$$y = \frac{1}{\ln 10} 2x \cdot \ln x^{\frac{1}{2}}$$

$$y = \frac{1}{\ln 10} \cancel{2x} \cdot \cancel{\frac{1}{2}} \ln x$$

$$y = \frac{1}{\ln 10} x \ln x$$

$$y' = \frac{1}{\ln 10} \left[(1) \ln x + x \frac{1}{x} \right]$$

اشتق بـ + بـ اشتق



Logarithmic differentiation



In

اسلوب يتم اللجوء له بالاشتقاق باضافة
والاستفادة من خواصها قبل الاشتقاق

يتم اللجوء له عند طلب اشتقاق

الدوال المعقدة

لوية بالبسط أو المقام

$$y = \frac{x^2(x-5)^3}{\sin^7 x \sqrt{\cos x}}$$

دالة
دالة

... $x^{\dots x}$

AL OTAYBI'S
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Find y'

$$y = x^x$$

$$\ln y = \ln x^x$$

$$\ln y = x \ln x$$

Diff. w.r.t. x

$$\frac{y'}{y} = (1) \ln x + x \cdot \frac{1}{x}$$

الشتقا

*y

$$y' = y [\ln x + 1]$$

$$= x^x [\ln x + 1]$$

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Let $f(x) = (3 + 4x)^{\cos x}$

July 2017

Find $f'(x)$.

$$\ln f(x) = \ln (3+4x)^{\cos x}$$

$$\ln f(x) = \cos x \ln (3+4x)$$

Diff. w.r.t. x

اشتق بـ x + اشتق

$$\frac{f'(x)}{f(x)} = -\sin x \ln(3+4x) + \cos x \cdot \frac{4}{3+4x}$$

* $f(x)$

$$f'(x) = f(x) \left[-\sin x \ln(3+4x) + \cos x \cdot \frac{4}{3+4x} \right]$$
$$= (3+4x)^{\cos x} \left[-\sin x \ln(3+4x) + \cos x \cdot \frac{4}{3+4x} \right]$$

Find $\frac{dy}{dx}$

$$y = \pi^{\cot x} x^{\tan^{-1} x}$$

$$\ln y = \ln [\pi^{\cot x} x^{\tan^{-1} x}]$$

$$\ln y = \ln \pi^{\cot x} + \ln x^{\tan^{-1} x}$$

$$\ln y = \cot x \ln \pi + \tan^{-1} x \ln x$$

Diff. w.r.t. x

$$\frac{y'}{y} = -\csc^2 x \ln \pi + \frac{1}{x^2+1} \ln x + \tan^{-1} x \cdot \frac{1}{x}$$

*y

$$y' = y \left[-\csc^2 x \ln \pi + \frac{1}{x^2+1} \ln x + \tan^{-1} x \cdot \frac{1}{x} \right]$$
$$= \pi^{\cot x} x^{\tan^{-1} x} \left[-\csc^2 x \ln \pi + \frac{1}{x^2+1} \ln x + \tan^{-1} x \cdot \frac{1}{x} \right]$$

Let $y = (x^2 + 1)^{\cos x}$. Use logarithmic differentiation

to find $\frac{dy}{dx}$ at $x = 0$

May 2017

$$y = (x^2 + 1)^{\cos x}$$

$$\ln y = \ln (x^2 + 1)^{\cos x}$$

$$\ln y = \cos x \ln (x^2 + 1)$$

Diff. w.r.t. x

اشتق بـ y + اشتق بـ $\ln$

$$\frac{y'}{y} = -\sin x \cdot \ln(x^2 + 1) + \cos x \cdot \frac{2x}{x^2 + 1}$$

* y

$$y' = y \left[-\sin x \cdot \ln(x^2 + 1) + \cos x \cdot \frac{2x}{x^2 + 1} \right]$$
$$= (x^2 + 1)^{\cos x} \left[-\sin x \ln(x^2 + 1) + \cos x \cdot \frac{2x}{x^2 + 1} \right]$$

at $x = 0$

$$y' = (0 + 1)^{\cos 0} [0 + (1)(0)]$$

$$= (1)(0) \rightarrow 0$$

$$y' = 0$$

Find y'

$$y = \frac{x^2(x-5)^3}{\sin^7 x \sqrt{\cos x}}$$

$$\ln y = \ln \frac{x^2(x-5)^3}{\sin^7 x (\cos x)^{\frac{1}{2}}}$$

$$\ln y = \ln x^2 + \ln(x-5)^3 - \ln \sin^7 x - \ln(\cos x)^{\frac{1}{2}}$$

$$\ln y = 2 \ln x + 3 \ln(x-5) - 7 \ln \sin x - \frac{1}{2} \ln \cos x$$

Diff. w.r.t. x

$$\frac{y'}{y} = 2 \frac{1}{x} + 3 \frac{1}{x-5} - 7 \frac{\cos x}{\sin x} - \frac{1}{2} \frac{-\sin x}{\cos x}$$

$\star y$

$$y' = y \left[\frac{2}{x} + \frac{3}{x-5} - 7 \cot x + \frac{1}{2} \tan x \right]$$

$$= \frac{x^2(x-5)^3}{\sin^7 x \sqrt{\cos x}} \left[\frac{2}{x} + \frac{3}{x-5} - 7 \cot x + \frac{1}{2} \tan x \right]$$

Find $\frac{dy}{dx}$ if $x^y = y^x$

Nov 2018

$$\ln x^y = \ln y^x$$

$$y \ln x = x \ln y$$

Diff. w.r.t. x

الشيء بـ y + مشتق y $\times$ مشتق $\ln x$ = (1) $\ln y$ + $x \times$ مشتق $\ln y$

$$y' \ln x + y \frac{1}{x} = (1) \ln y + x \frac{y'}{y}$$

$$y' \ln x - x \frac{y'}{y} = \ln y - y \frac{1}{x}$$

$$y' \left[\ln x - \frac{x}{y} \right] = \ln y - \frac{y}{x}$$

$$y' = \frac{\ln y - \frac{y}{x}}{\ln x - \frac{x}{y}}$$

Use logarithmic differentiation to find y'

$$y = (x^2 + 2)^2 (x^4 + 4)^4$$

$$\ln y = \ln [(x^2 + 2)^2 (x^4 + 4)^4]$$

$$\ln y = \ln (x^2 + 2)^2 + \ln (x^4 + 4)^4$$

$$\ln y = 2 \ln (x^2 + 2) + 4 \ln (x^4 + 4)$$

Diff. w.r.t. x

$$\begin{aligned} \frac{y'}{y} &= 2 \frac{2x}{x^2 + 2} + 4 \frac{4x^3}{x^4 + 4} \\ * y \quad y' &= y \left[2 \frac{2x}{x^2 + 2} + 4 \frac{4x^3}{x^4 + 4} \right] \\ y' &= (x^2 + 2)^2 (x^4 + 4)^4 \left[2 \frac{2x}{x^2 + 2} + 4 \frac{4x^3}{x^4 + 4} \right] \end{aligned}$$

Use logarithmic differentiation to find y'

$$y = \sqrt{\frac{x-1}{x^4+1}}$$

$$\ln y = \ln \left(\frac{x-1}{x^4+1} \right)^{\frac{1}{2}}$$

$$\ln y = \frac{1}{2} \ln \left(\frac{x-1}{x^4+1} \right)$$

$$\ln y = \frac{1}{2} \left[\ln(x-1) - \ln(x^4+1) \right]$$

Diff. w.r.t. x

$$\begin{aligned} * y \quad \frac{y'}{y} &= \frac{1}{2} \left[\frac{1}{x-1} - \frac{4x^3}{x^4+1} \right] \\ y' &= y \frac{1}{2} \left[\frac{1}{x-1} - \frac{4x^3}{x^4+1} \right] \\ &= \sqrt{\frac{x-1}{x^4+1}} \frac{1}{2} \left[\frac{1}{x-1} - \frac{4x^3}{x^4+1} \right] \end{aligned}$$

Use logarithmic differentiation to find y'

$$y = (\cos x)^x$$

$$\ln y = \ln (\cos x)^x$$

$$\ln y = x \ln \cos x$$

Diff. w.r.t. x

الشتق بشتا + بشتا الشقا

$$\frac{y'}{y} = (1) \ln \cos x + x \frac{-\sin x}{\cos x}$$

* y

$$\begin{aligned} y' &= y [\ln \cos x + x (-\tan x)] \\ &= (\cos x)^x [\ln \cos x + x (-\tan x)] \end{aligned}$$

Use logarithmic differentiation to find y'

$$y = (\ln x)^{\cos x}$$

$$\ln y = \ln (\ln x)^{\cos x}$$

$$\ln y = (\cos x) \ln (\ln x)$$

Diff. w.r.t. x

الشتق بشتا + بشتا الشقا

$$\frac{y'}{y} = -\sin x \ln (\ln x) + \cos x \frac{\frac{1}{x}}{\ln x}$$

* y

$$y' = y \left[-\sin x \ln (\ln x) + \cos x \frac{\frac{1}{x}}{\ln x} \right]$$

$$= (\ln x)^{\cos x} \left[-\sin x \ln (\ln x) + \cos x \frac{\frac{1}{x}}{\ln x} \right]$$

Find an equation of the tangent line to the curve

$y = x^{\ln x}$ at the point $(1, 1)$

Dec 2016

$$y = x^{\ln x}$$

$$\ln y = \ln x^{\ln x}$$

$$\ln y = \ln x \ln x \\ = (\ln x)^2$$

Diff. w.r.t. x

$$\frac{y'}{y} = 2(\ln x) \cdot \frac{1}{x}$$

* y

$$y' = y \cdot 2(\ln x) \cdot \frac{1}{x}$$

at $y' = x^{\ln x} \cdot 2(\ln x) \cdot \frac{1}{x}$

P(1,1)

$$y' = 1^{\ln 1} \cdot 2(\ln 1) \cdot \frac{1}{1}$$

$$= 0$$

∴ The equation of the tangent line

$$y - y_0 = m(x - x_0)$$

$$y - 1 = 0$$

Let $f(x) = \log_{10}(x^2 + 1)$ Then : is

a) $f'(x) = \frac{2x}{x^2 + 1}$ b) $f'(x) = \frac{2x}{(x^2 + 1)\ln 10}$

c) $f'(x) = \frac{1}{x^2 + 1}$ d) $f'(x) = \frac{1}{(x^2 + 1)\ln 10}$

e) None of the above.

Dec 2017

$f(x) = \frac{1}{\ln 10} \ln(x^2 + 1)$

$f'(x) = \frac{1}{\ln 10} \cdot \frac{2x}{x^2 + 1}$ (b) ✓

The derivative of $f(x) = x^{\sqrt{x+1}}$ at $x = 1$ is

a) 0 b) 1 c) $\sqrt{2}$ d) 2 e) None of the above.

Apr 2019

$f(x) = x^{\sqrt{x+1}}$

$\ln f(x) = \ln x^{\sqrt{x+1}}$

Diff. w.r.t. x $\ln f(x) = \sqrt{x+1} \ln x$

$\frac{f'(x)}{f(x)} = \frac{1}{2\sqrt{x}} \ln x + \sqrt{x+1} \cdot \frac{1}{x}$

$f'(x) = f(x) \left[\frac{1}{2\sqrt{x}} \ln x + \sqrt{x+1} \cdot \frac{1}{x} \right]$

$= x^{\sqrt{x+1}} \left[\frac{1}{2\sqrt{x}} \ln x + \sqrt{x+1} \cdot \frac{1}{x} \right]$

$f'(1) = 1^{\sqrt{2}} [0 + \sqrt{2} \cdot \frac{1}{1}] = \sqrt{2}$ (c) ✓

Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$$

$\frac{0}{0}$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f(x) = \ln x$$

$$a = 1$$

$$f'(x) = \frac{1}{x}$$

$$f'(a) = \frac{1}{1} = 1$$

$$\therefore \lim_{x \rightarrow 1} \frac{\ln x}{x - 1} = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{3^x - 1}{x}$$

$\frac{0}{0}$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f(x) = 3^x$$

$$a = 0$$

$$f'(x) = 3^x (1) \ln 3$$

$$f'(a) = 3^0 \ln 3 = (1) \ln 3 = \ln 3$$

$$\therefore \lim_{x \rightarrow 0} \frac{3^x - 1}{x} = \ln 3$$

The value of

Nov 2018

$$\lim_{x \rightarrow 0} \frac{e^{x^2+3x} - 1}{x} =$$

a) 2 b) 3 c) 6 d) Does not exist e) None of the above

$$\lim_{x \rightarrow 0} \frac{e^{x^2+3x} - 1}{x} = \frac{1-1}{0} = \frac{0}{0}$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f(x) = e^{x^2+3x}$$

$$a = 0$$

$$f'(x) = e^{x^2+3x} (2x+3)$$

$$f(a) = e^0 (0+3) = (1)(3) = 3$$

$$\therefore \lim_{x \rightarrow 0} \frac{e^{x^2+3x} - 1}{x} = 3 \quad (b)$$

* قراءة *

Find y'

$$y = \left[\frac{4 - \tan^{-1} x^2}{e^x \sqrt{\tan x}} \right]^x$$

$$\ln y = \ln \left[\frac{4 - \tan^{-1} x^2}{e^x (\tan x)^{\frac{1}{2}}} \right]^x$$

$$\ln y = x \ln \left[\frac{4 - \tan^{-1} x^2}{e^x (\tan x)^{\frac{1}{2}}} \right]$$

$$\ln y = x \left[\ln(4 - \tan^{-1} x^2) - \ln e^x - \ln(\tan x)^{\frac{1}{2}} \right]$$

$$\ln y = x \left[\ln(4 - \tan^{-1} x^2) - x - \frac{1}{2} \ln \tan x \right]$$

Diff. w.r.t. x

$$\frac{y'}{y} = (1) \left[\ln(4 - \tan^{-1} x^2) - x - \frac{1}{2} \ln \tan x \right]$$

$$+ x \left[\frac{0 - \frac{2x}{(x^2)^2 + 1}}{4 - \tan^{-1} x^2} - 1 - \frac{1}{2} \frac{\sec^2 x}{\tan x} \right]$$

* y

$$y' = y \left[\text{انزله كله بالقوس} \right]$$

$$= \left[\frac{4 - \tan^{-1} x^2}{e^x \sqrt{\tan x}} \right]^x \left[\text{انزله كله بالقوس} \right]$$

Find y'

$$y = \frac{\sin^8 x (x^7 - 5)^5}{\sqrt{x^2 - 3}}$$

$$\ln y = \ln \frac{\sin^8 x (x^7 - 5)^5}{(x^2 - 3)^{\frac{1}{2}}}$$

$$\ln y = \ln \sin^8 x + \ln (x^7 - 5)^5 - \ln (x^2 - 3)^{\frac{1}{2}}$$

$$\ln y = 8 \ln \sin x + 5 \ln (x^7 - 5) - \frac{1}{2} \ln (x^2 - 3)$$

Diff. w.r.t. x

$$\frac{y'}{y} = 8 \frac{\cos x}{\sin x} + 5 \frac{7x^6}{x^7 - 5} - \frac{1}{2} \frac{2x}{x^2 - 3}$$

*y

$$y' = y \left[8 \cot x + \frac{35x^6}{x^7 - 5} - \frac{x}{x^2 - 3} \right]$$

$$= \frac{\sin^8 x (x^7 - 5)}{\sqrt{x^2 - 3}} \left[8 \cot x + \frac{35x^6}{x^7 - 5} - \frac{x}{x^2 - 3} \right]$$

Find $\frac{dy}{dx}$

$$y = (\cos x)^{\sin x} + \ln x + e^{\pi}$$



إذا كانت الدالة المعقدة مجموعة أو مطروحة
يتم أخذها لوحدها وتسميتها وإضافة $\ln$ عليها لوحدها
واشتقاق كل دالة بشكل منفصل عن الأخرى



مهمبة !! لو تمت إحنافه $\ln$ للسؤال بأكمه

Let $u = (\cos x)^{\sin x}$

$$\ln u = \ln(\cos x)^{\sin x}$$

$$\ln u = \sin x \ln \cos x$$

Diff. w.r.t. x

$$\frac{u'}{u} = \cos x \ln \cos x + \sin x \frac{-\sin x}{\cos x}$$

* u

$$u' = u \left[\cos x \ln \cos x + \sin x \frac{-\sin x}{\cos x} \right]$$

$$= (\cos x)^{\sin x} \left[\cos x \ln \cos x + \sin x \frac{-\sin x}{\cos x} \right]$$

$$\Rightarrow y' = (\cos x)^{\sin x} \left[\cos x \ln \cos x + \sin x \frac{-\sin x}{\cos x} \right]$$

$+ \frac{1}{x} + 0$
 $\ln x$ مسته ← (رقم) ← مسته



Tricky question



Find $\frac{d}{dx} [f(\ln x) + [f(x)]^{\sin x}]$

$f'(\ln x) \rightarrow \frac{1}{x}$

Let $u = [f(x)]^{\sin x}$

$\ln u = \ln [f(x)]^{\sin x}$

$\ln u = \sin x \ln f(x)$

Diff w.r.t. x

$\frac{u'}{u} = \cos x \ln f(x) + \sin x \frac{f'(x)}{f(x)}$

$\times u$

$u' = u [\cos x \ln f(x) + \sin x \frac{f'(x)}{f(x)}]$
 $= [f(x)]^{\sin x} [\cos x \ln f(x) + \sin x \frac{f'(x)}{f(x)}]$

$\Rightarrow f'(\ln x) \frac{1}{x} + [f(x)]^{\sin x} [\cos x \ln f(x) + \sin x \frac{f'(x)}{f(x)}]$

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